

Cells evolved as single, free-living organisms, but natural selection favoured more complex communities of cells, multicellular organisms, where groups of cells specialize during development to carry out specific functions for the body as a whole. This allowed the emergence of larger organisms with greater control over their internal environment and the evolution of highly specialized organic structures such as the brain. The human body contains more than 200 different cell types, sharing the same genome but with different patterns of gene expression.

Some cells in the body are essentially migratory, but most exist as cellular aggregates in which individual cells carry out similar or closely related functions in a coordinated manner. These aggregates are termed tissues, and can be classified into a fairly small number of broad categories on the basis of their structure, function and molecular properties. On the basis of their structure, most tissues are divided into four major types: epithelia, connective or supporting tissue, muscle and nervous tissue. Epithelia are continuous layers of cells with little intercellular space, which cover or line surfaces, or have been so derived. In connective tissues, the cells are embedded in an extracellular matrix, which, typically, forms a substantial and important component of the tissue. Muscle consists largely of specialized contractile cells. Nervous tissue consists of cells specialized for conducting and transmitting electrical and chemical signals and the cells that support this activity.

There is molecular evidence that this structure-based scheme of classification has validity. Thus the intermediate filament proteins characteristic of all epithelia are keratins (Pan et al 2012); those of connective tissue are vimentins; those of muscle are desmins; and those of nervous tissue are neurofilament and glial fibrillary acidic proteins. However, cells such as myofibroblasts, neuroepithelial sensory receptors and ependymal cells of the central nervous system have features of more than one tissue type. Despite its anomalies, the scheme is useful for descriptive purposes; it is widely used and will be adopted here.

In this chapter, two of the major tissue categories, epithelia and general connective and supporting tissues, will be described. Specialized skeletal connective tissues, i.e. cartilage and bone, together with skeletal muscle, are described in detail in Chapter 5 as part of the musculoskeletal system overview. Smooth muscle and cardiac muscle are described in Chapter 6. Nervous system tissues are described in Chapter 3. Specialized defensive cells, which also form a migrant population within the general connective tissues, are considered in more detail in Chapter 4, with blood, lymphoid tissues and haemopoiesis.

## EPITHELIA

The term epithelium is applied to the layer or layers of cells that cover the body surfaces or line the body cavities that open on to it. The fate of embryonic epithelial populations is illustrated in Figure 12.3. Epithelia function generally as selective barriers that facilitate, or inhibit, the passage of substances across the surfaces they cover. In addition, they may: protect underlying tissues against dehydration, chemical or mechanical damage; synthesize and secrete products into the spaces that they line; and function as sensory surfaces. In this respect, many features of nervous tissue can be regarded as those of a modified epithelium and the two tissue types share an origin in embryonic ectoderm.

Epithelia (Fig. 2.1) are predominantly cellular and the little extracellular material they possess is limited to the basal lamina. Intercellular junctions, which are usually numerous, maintain the mechanical cohesiveness of the epithelial sheet and contribute to its barrier functions. A series of three intercellular junctions forms a typical epithelial junctional complex: in sequence from the apical surface, this consists of a tight junctional zone, an adherent (intermediate) junctional zone and a region of discrete desmosome junctions. Epithelial cell shape is most usually polygonal and partly determined by cytoplasmic features such as secretory granules. The basal surface of an epithelium lies in contact with a thin layer of filamentous protein and proteoglycan termed the

basal lamina, which is synthesized predominantly by the epithelial cells. The basal lamina is described on page 34.

Epithelia can usually regenerate when injured. Indeed, many epithelia continuously replace their cells to offset cell loss caused by mechanical abrasion (reviewed in Blanpain et al (2007)). Blood vessels do not penetrate typical epithelia and so cells receive their nutrition by diffusion from capillaries of neighbouring connective tissues. This arrangement limits the maximum thickness of living epithelial cell layers. Epithelia, together with their supporting connective tissue, can often be removed surgically as one layer, which is collectively known as a membrane. Where the surface of a membrane is moistened by mucous glands it is called a mucous membrane or mucosa, whereas a similar layer of connective tissue covered by mesothelium is called a serous membrane or serosa.

## CLASSIFICATION

Epithelia can be classified as unilaminar (single-layered, simple), in which a single layer of cells rests on a basal lamina; or multilaminar, in which the layer is more than one cell thick. The latter includes: stratified squamous epithelia, in which flattened superficial cells are constantly replaced from the basal layers; urothelium (transitional epithelium), which serves special functions in the urinary tract; and other multilaminar epithelia such as those lining the largest ducts of some exocrine glands, which, like urothelium, are replaced only very slowly under normal conditions. Seminiferous epithelium is a specialized multilaminar tissue found only in the testis.

## Unilaminar (simple) epithelia

Unilaminar epithelia are further classified according to the shape of their cells, into squamous, cuboidal, columnar and pseudostratified types. Cell shape may, in some cases, be related to cell volume. Where little cytoplasm is present, there are generally few organelles and therefore there is low metabolic activity and cells are squamous or low cuboidal. Highly active cells, e.g. secretory epithelia, contain abundant mitochondria and endoplasmic reticulum, and are typically tall cuboidal or columnar. Unilaminar epithelia can also be subdivided into those that have special functions, such as those with cilia, numerous microvilli, secretory vacuoles (in mucous and serous glandular cells) or sensory features. Myoepithelial cells, which are contractile, are found as isolated cells associated with glandular structures, e.g. salivary and mammary glands.

### Squamous epithelium

Simple squamous epithelium is composed of flattened, tightly apposed, polygonal cells (squames). This type of epithelium is described as tessellated when the cells have complex, interlocking borders rather than straight boundaries. The cytoplasm may in places be only 0.1 µm thick and the nucleus usually bulges into the overlying space (Fig. 2.2A). These cells line the alveoli of the lungs, where their surface area is huge and cytoplasmic volume correspondingly large, and they also form the outer capsular wall of renal corpuscles, the thin segments of the renal tubules and various parts of the inner ear. Because it is so thin, simple squamous epithelium allows rapid diffusion of gases and water across its surface; it may also engage in active transport, as indicated by the presence of numerous endocytic vesicles in these cells. Tight junctions (occluding junctions, zonulae adherentes) between adjacent cells ensure that materials pass primarily through cells, rather than between them.

### Cuboidal and columnar epithelia

Cuboidal and columnar epithelia consist of regular rows of cylindrical cells (Figs 2.2B, C). Cuboidal cells are approximately square in vertical section, whereas columnar cells are taller than their diameter, and both

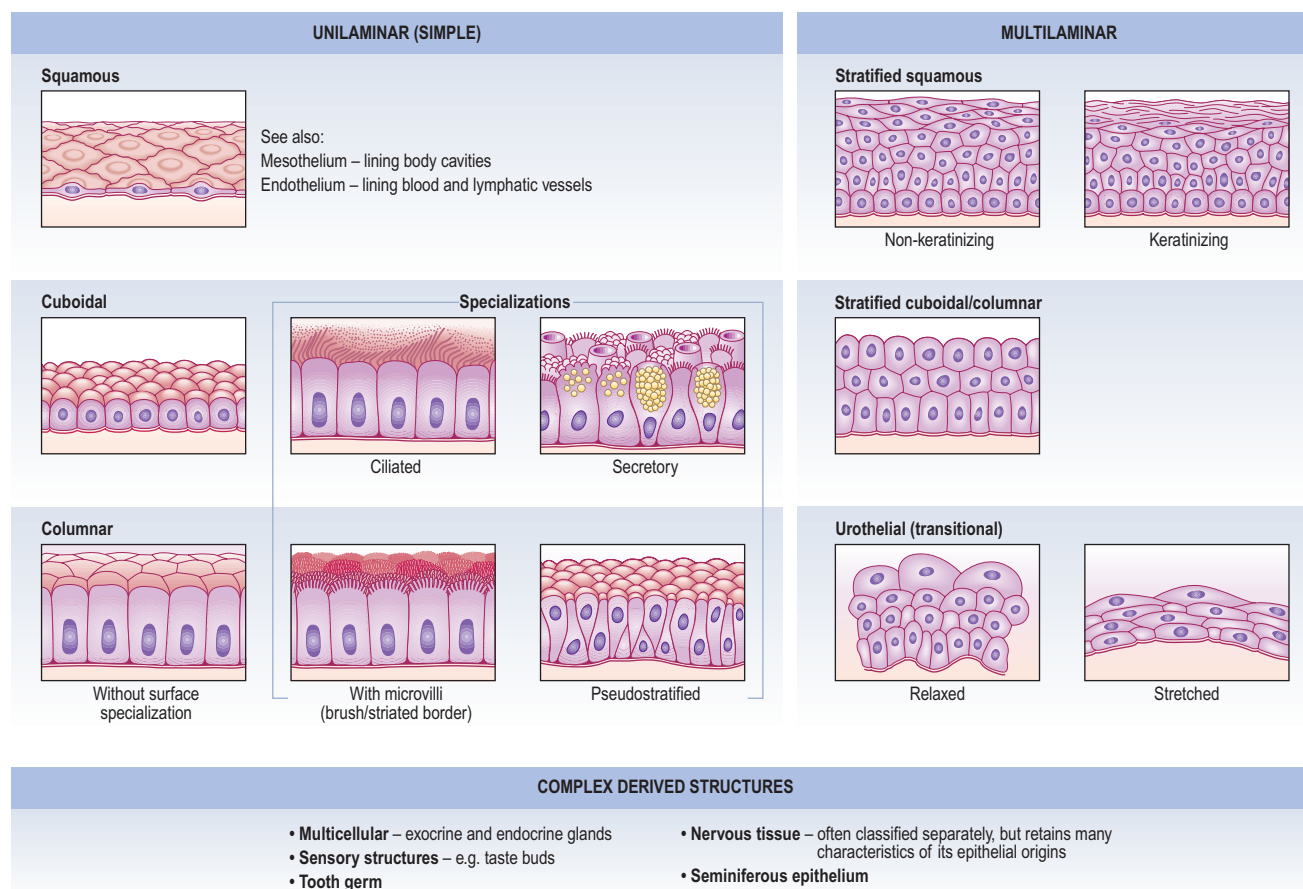


Fig. 2.1 Classification of epithelial tissues and cells.

are polygonal when sectioned horizontally. Commonly, microvilli are found on their free surfaces, which considerably increases the absorptive area, e.g. in the epithelia of the small intestine (columnar cells with a striated border of very regular microvilli), the gallbladder (columnar cells with a brush border of microvilli); proximal convoluted tubules of the kidney (large cuboidal to low columnar cells with brush borders); and the epididymis (columnar cells with extremely long microvilli, erroneously termed stereocilia).

Ciliated columnar epithelium lines most of the respiratory tract, except for the lower pharynx and vocal folds, and it is pseudostratified (Fig. 2.2D) as far as the larger bronchioles; it also lines some of the tympanic cavity and auditory tube; the uterine tube; and the efferent ductules of the testis. Submucosal mucous glands and mucosal goblet cells secrete mucus on to the luminal surface of much of the respiratory tract, and cilia sweep a layer of mucus, trapped dust particles and so on from the lung towards the pharynx in the mucociliary rejection current, which clears the respiratory passages of inhaled particles. Cilia in the uterine tube assist the passage of oocytes and fertilized ova to the uterus.

Some columnar cells are specialized for secretion, and aggregates of such cells may be described as glandular tissue. Their apical domains typically contain mucus- or protein-filled (zymogen) vesicles, e.g. mucin-secreting and chief cells of the gastric epithelium. Where mucous cells lie among non-secretory cells, e.g. in the intestinal epithelium, their apical cytoplasm and its secretory contents often expand to produce a characteristic cell shape, and they are known as goblet cells (see Fig. 2.2D). For further details of glandular tissue, see page 32, and for the characteristics of mucus, see page 40.

### Pseudostratified epithelium

Pseudostratified epithelium is a single-layered (simple) columnar epithelium in which nuclei lie at different levels in a vertical section (Fig. 2.2D). All cells are in contact with the basal lamina throughout their lifespan, but not all cells extend through the entire thickness of the epithelium. Some constitute an immature basal cell layer of smaller cells, which are often mitotic and able to replace damaged mature cells. Migrating lymphocytes and mast cells within columnar epithelia may also give a similar, pseudostratified appearance because their nuclei are found at different depths. Much of the ciliated lining of the respiratory tract is of the pseudostratified type, and so is the sensory epithelium of the olfactory area.

### Sensory epithelia

Sensory epithelia are found in special sense organs of the olfactory, gustatory and vestibulocochlear receptor systems. All of these contain sensory cells surrounded by supportive non-receptor cells. Olfactory receptors are modified neurones and their axons pass directly to the brain, but the other types are specialized epithelial cells that synapse with terminals of afferent (and sometimes efferent) nerve fibres.

### Myoepithelial cells

Myoepithelial cells, which are also sometimes termed basket cells, are fusiform or stellate in shape (Fig. 2.3), contain actin and myosin filaments, and contract when stimulated by nervous or endocrine signals. They surround the secretory portions and ducts of some glands, e.g. mammary, lacrimal, salivary and sweat glands, and lie between the basal lamina and the glandular or ductal epithelium. Their contraction assists the initial flow of secretion into larger conduits. Myoepithelial cells are ultrastructurally similar to smooth muscle cells in the arrangement of their actin and myosin, but differ from them because they originate, like the glandular cells, from embryonic ectoderm or endoderm. They can be identified immunohistochemically on the basis of the co-localization of myofilament proteins (which signify their contractile function (Fig. 2.4)) and keratin intermediate filaments (which accords with their epithelial lineage).

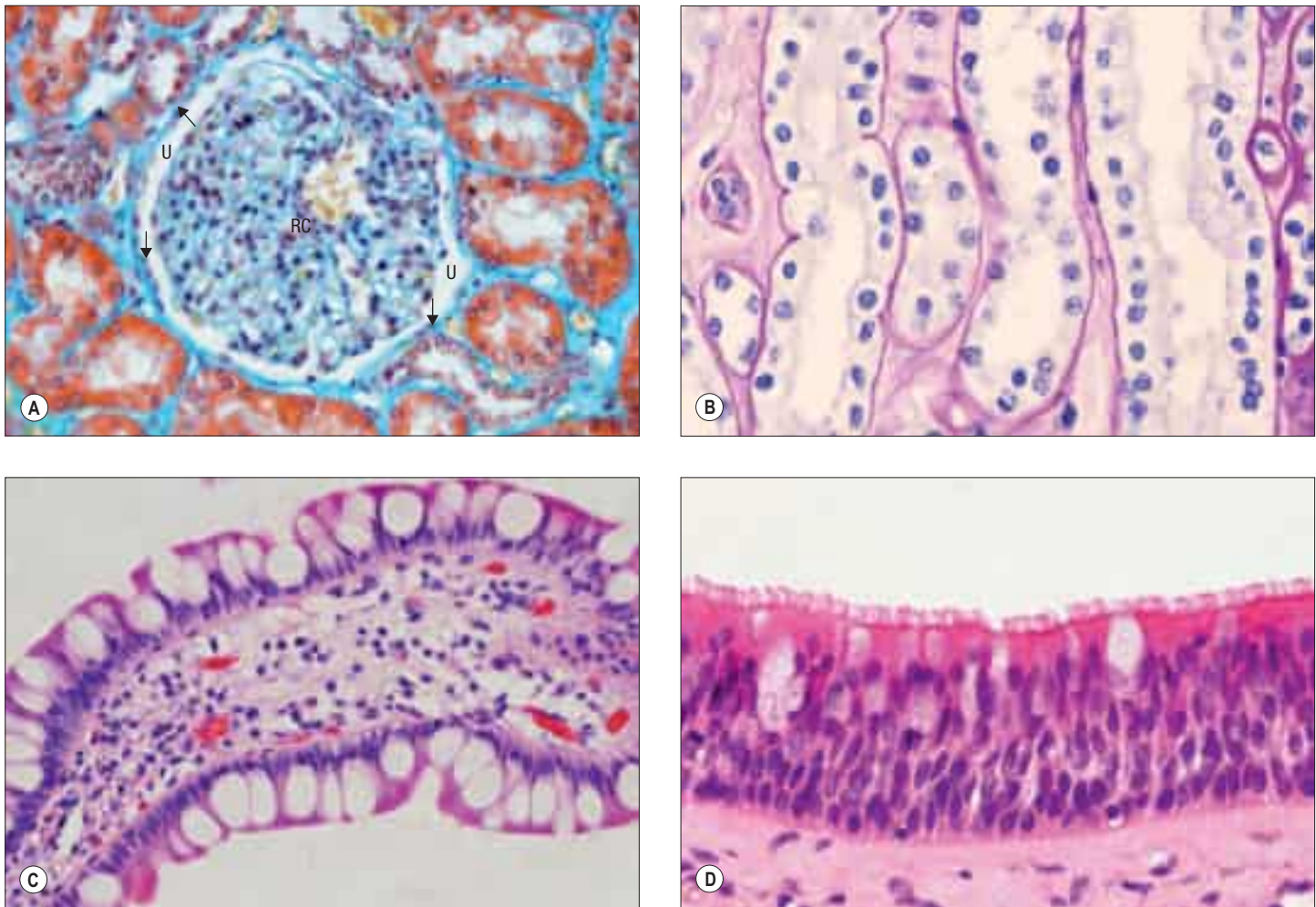
### Multilaminar (stratified) epithelia

Multilaminar epithelia are found at surfaces subjected to mechanical damage or other potentially harmful conditions. They can be divided into those that continue to replace their surface cells from deeper layers, designated stratified squamous epithelia, and others in which replacement is extremely slow except after injury.

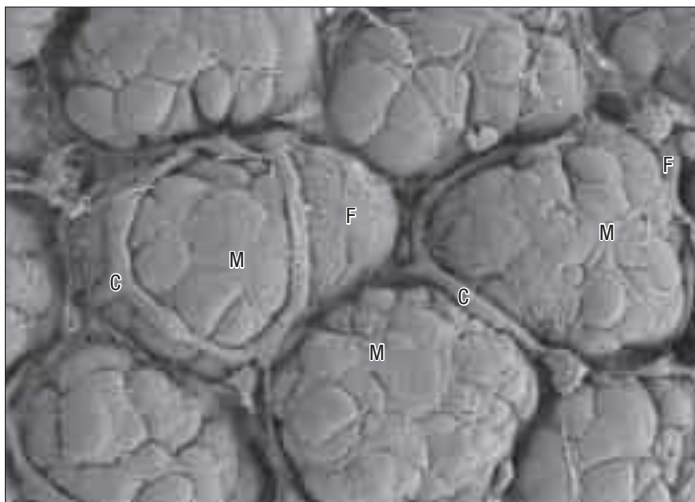
### Stratified squamous epithelia

Stratified squamous epithelia are multilayered tissues in which the formation, maturation and loss of cells is continuous, although the rates of these processes can change, e.g. after injury. New cells are formed in the most basal layers by the mitotic division of stem cells and transit (or transient) amplifying cells. The daughter cells move more superficially, changing gradually from a cuboidal shape to a more

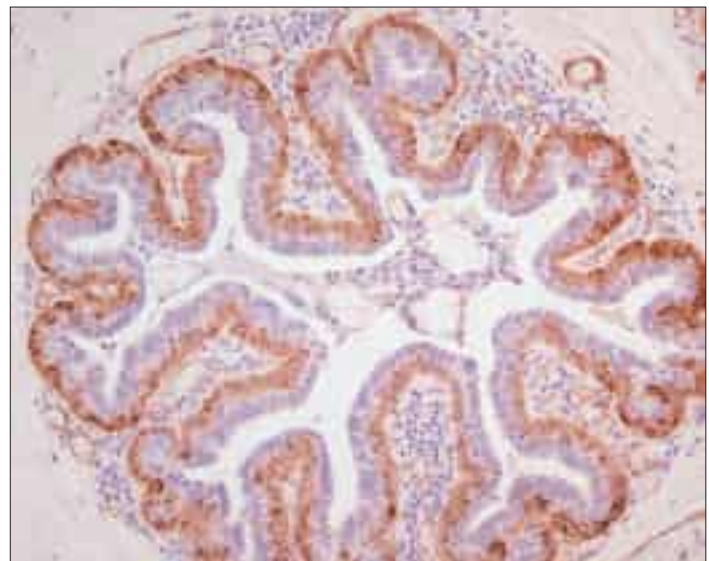




**Fig. 2.2** **A**, Simple squamous epithelium lining the outer parietal layer (arrows) of a Bowman's capsule in the renal corpuscle (RC), stained with the trichrome, Martius Scarlet Blue (MSB). Oval epithelial nuclei project into the urinary space (U), within a highly attenuated cytoplasm. **B**, Simple cuboidal epithelium lining a group of collecting ducts sectioned longitudinally in the renal medulla. The basement membranes are stained magenta with periodic-acid Schiff (PAS) reagent. **C**, Simple columnar epithelium covering the tip (off field, right) of a villus in the ileum. Tall, columnar absorptive cells with oval, vertically orientated nuclei bear a striated border of microvilli, just visible as a deeper-stained apical fringe. Numerous interspersed goblet cells are present, with pale apical cytoplasm filled with mucinogen secretory granules and dark, flattened, basally situated nuclei. **D**, Ciliated columnar pseudostratified epithelium in the respiratory tract, and interspersed goblet cells, with pale, mucinogen granule-filled apical cytoplasm. All human tissues. (All human tissues, courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

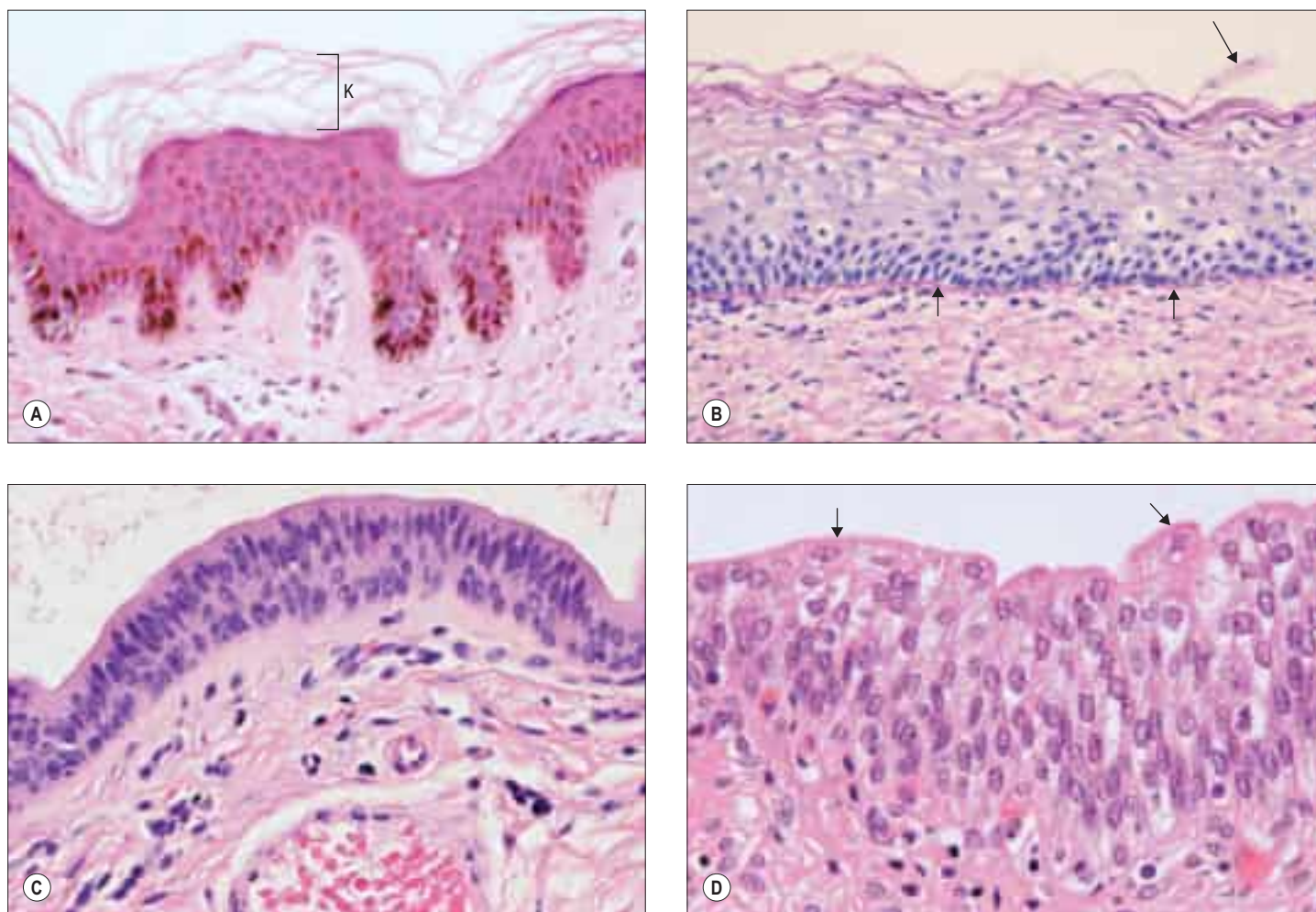


**Fig. 2.3** Stellate myoepithelial cells (M) wrapped around secretory acini in the lactating mouse mammary gland, seen in the scanning electron microscope after enzymatic depletion of extracellular matrix. Blood capillaries (C) and fibroblasts (F) are also indicated. (Courtesy of Prof. Toshikazu Nagato, Fukuoka Dental College, Japan.)



**Fig. 2.4** Myoepithelial cells (stained brown), in a human breast duct, demonstrated immunohistochemically using antibody to smooth muscle actin. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)





**Fig. 2.5** **A**, Keratinized stratified squamous epithelium in thin skin. Pigmented melanocytes are seen in the basal layer and a few keratinocytes of the prickly cell layer also contain melanin granules. The dead, keratinized layer (K) lacks nuclei. **B**, Non-keratinized stratified squamous epithelium of the uterine ectocervix, stained with periodic-acid Schiff (PAS) reagent. The basement membrane (short arrows) and superficial squames, which retain their nuclei, are PAS-positive; squames sloughing off the surface are indicated (long arrow). **C**, Stratified low columnar epithelium of an interlobular excretory duct of the sublingual salivary gland. **D**, Urothelium (transitional epithelium) lining the relaxed urinary bladder. The most superficial cells have a thickened plasma membrane as a result of the presence of intramembranous plaques, which give an eosinophilic appearance to the luminal surface (arrows). All human tissues. (All human tissues, courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

flattened form, and are eventually shed from the surface as a highly flattened squame. Typically, the cells are held together by numerous desmosomes to form strong, contiguous cellular sheets that provide protection to the underlying tissues against mechanical, microbial and chemical damage. Stratified squamous epithelia may be broadly subdivided into keratinized and non-keratinized types.

### Keratinized epithelium

Keratinized epithelium (**Fig. 2.5A**) is found at surfaces that are subject to drying or mechanical stresses, or are exposed to high levels of abrasion. These include the entire epidermis and the mucocutaneous junctions of the lips, nostrils, distal anal canal, outer surface of the tympanic membrane and parts of the oral lining (gingivae, hard palate and filiform papillae on the anterior part of the dorsal surface of the tongue). Their cells, keratinocytes, are described in more detail on [page 141](#). A distinguishing feature of keratinized epithelia is that cells of the superficial layer, the stratum corneum, are anucleate, dead, flattened squames that eventually flake off from the surface. In addition, the tough keratin intermediate filaments become firmly embedded in a matrix protein. This unusual combination of strongly coherent layers of living cells and more superficial strata made of plates of inert, mechanically robust protein complexes, interleaved with water-resistant lipid, makes this type of epithelium an efficient barrier against different types of injury, microbial invasion and water loss.

### Non-keratinized epithelium

Non-keratinized epithelium is present at surfaces that are subject to abrasion but protected from drying (**Fig. 2.5B**). These include: the buccal cavity (except for the areas noted above); oropharynx and laryngopharynx; oesophagus; part of the anal canal; vagina; distal uterine cervix; distal urethra; cornea; inner surfaces of the eyelids; and the vestibule of the nasal cavities. Cells go through the same transitions in

general shape as are seen in the keratinized type, but they do not fill completely with keratin or secrete glycolipid, and they retain their nuclei until they desquamate at the surface. In sites where considerable abrasion occurs, e.g. parts of the buccal cavity, the epithelium is thicker and its most superficial cells may partly keratinize, so that it is referred to as parakeratinized, in contrast to the orthokeratinized state of fully keratinized epithelium. Diets deficient in vitamin A may induce keratinization of such epithelia, and excessive doses may lead to its transformation into mucus-secreting epithelium.

### Stratified cuboidal and columnar epithelia

Two or more layers of cuboidal or low columnar cells (**Fig. 2.5C**) are typical of the walls of the larger ducts of some exocrine glands, e.g. the pancreas, salivary glands and the ducts of sweat glands, and they presumably provide more strength than a single layer. Parts of the male urethra are also lined by stratified columnar epithelium. The layers are not continually replaced by basal mitoses and there is no progression of form from base to surface, but they can repair themselves if damaged.

### Urothelium (urinary or transitional epithelium)

Urothelium (**Fig. 2.5D**) is a specialized epithelium that lines much of the urinary tract and prevents its rather toxic contents from damaging surrounding structures. It extends from the ends of the collecting ducts of the kidneys, through the ureters and bladder, to the proximal portion of the urethra. In males it lines the urethra as far as the ejaculatory ducts, then becomes intermittent and is finally replaced by stratified columnar epithelium in the membranous urethra. In females it extends as far as the urogenital membrane.

The epithelium appears to be 4–6 cells thick and lines organs that undergo considerable distension and contraction. It can therefore stretch greatly without losing its integrity. In stretching, the cells become

## INTEGRATING CELLS INTO TISSUES

flattened without altering their positions relative to each other, because they are firmly connected by numerous desmosomes. However, the urothelium appears to be reduced to only 2–3 cells thick. The epithelium is called transitional because of the apparent transition from a stratified cuboidal epithelium to a stratified squamous epithelium, which occurs as it is stretched to accommodate urine, particularly in the bladder. The basal cells are basophilic and contain many ribosomes; they are uninucleate (diploid), and cuboidal when relaxed. More apically, they form large binucleate or, more often, polyploid uninucleate cells. The surface cells are the largest and may even be octoploid; in the relaxed state they typically bulge into the lumen as dome-shaped cells with a thickened, eosinophilic glycocalyx or cell coat. Their luminal surfaces are covered by a specialized plasma membrane in which plaques of intramembranous glycoprotein particles are embedded to stiffen the membrane. When the epithelium is relaxed, the surface area of the cells is reduced and the plaques are partially internalized by the hinge-like action of the more flexible interplaque membrane regions. The plaques re-emerge on to the surface when it is stretched.

Normally, cell turnover is very slow; cell division is infrequent and is restricted to the basal layer. However, when damaged, the epithelium regenerates quite rapidly.

### Seminiferous epithelium

Seminiferous epithelium is a highly specialized, complex stratified epithelium. It consists of a heterogeneous population of cells that form the lineage of the spermatozoa (spermatogonia, spermatocytes, spermatids), together with supporting cells (Sertoli cells). It is described in detail on [page 1275](#).

## GLANDS

One of the features of many epithelia is their ability to alter the environment facing their free surfaces by the directed transport of ions, water or macromolecules. This is particularly well demonstrated in glandular tissue, in which the metabolism and structural organization of the cells are specialized for the synthesis and secretion of macromolecules, usually from the apical surface. Such cells may exist in isolation amongst other non-secretory cells of an epithelium, e.g. goblet cells in the absorptive lining of the small intestine, or may form highly coherent sheets of epithelium with a common secretory function, e.g. the mucous lining of the stomach and, in a highly invaginated structure, the complex salivary glands.

Glands may be subdivided into exocrine glands and endocrine glands. Exocrine glands secrete, usually via a duct, on to surfaces that are continuous with the exterior of the body, including the alimentary tract, respiratory system, urinary and genital ducts and their derivatives, and the skin. Endocrine glands are ductless and secrete hormones directly into interstitial fluid and thence the circulatory system, which conveys them throughout the body to affect the activities of other cells. In addition to strictly epithelial glands, some tissues derived from the nervous system, including the suprarenal medulla and neurohypophysis, are neurosecretory.

Paracrine glandular cells are similar to endocrine cells but their secretions diffuse locally to cellular targets in the immediate vicinity; many are classed as neuroendocrine cells because they secrete molecules used elsewhere in the nervous system as neurotransmitters or neuromodulators. Modes of signalling by secretory cells are illustrated in [Figure 1.6](#).

## EXOCRINE GLANDS

### Types of secretory process

The mechanism of secretion varies considerably. If the secretions are initially packaged into membrane-bound vesicles, these are conveyed to the cell surface, where they are discharged. In merocrine secretion, which is by far the most common secretory mechanism, vesicle membranes fuse with the plasma membrane to release their contents to the exterior ([Fig. 2.6](#)). Specialized transmembrane molecules in the secretory vesicle wall recognize marker proteins on the cytoplasmic side of the plasma membrane and bind to them. This initiates interactions with other proteins that cause the fusion of the two membranes and the consequent release of the vesicle contents. The stimulus for secretion varies with the type of cell but often appears to involve a rise in intracellular calcium. Glands such as the simple sweat glands of the skin, where ions and water are actively transported from plasma as an exudate, were once classified as eccrine glands. They are now known to synthesize and

secrete small amounts of protein by a merocrine mechanism, and have been reclassified as merocrine glands.

In apocrine glands, some of the apical cytoplasm is pinched off with the contained secretions, which are stored in the cell as membrane-free droplets (see [Fig. 2.6](#)). The best-understood example of this is the secretion of milk fat by mammary gland cells, in which a small amount of cytoplasm is incorporated into the plasma membrane-bound lipid globule as it is released from the cell. Larger amounts of cytoplasm are included in secretions by specialized apocrine sweat glands in the axilla ([Stoeckelhuber et al 2011](#)) and anogenital regions of the body. In some tissues there is a combination of different types of secretion, e.g. mammary gland cells secrete milk fat by apocrine secretion and milk protein, casein, by merocrine secretion.

In holocrine glands (see [Fig. 2.6](#)), e.g. sebaceous glands in the skin, the cells first fill with secretory products (lipid droplets or sebum, in this instance), after which the entire cell disintegrates to liberate the accumulated mass of secretion into the adjacent duct or, more usually, hair follicle.

## Structural and functional classification

Exocrine glands are either unicellular or multicellular. The latter may be in the form of simple sheets of secretory cells, e.g. the lining of the stomach, or may be structurally more complex and invaginated to a variable degree. Such glands (see [Fig. 2.6](#)) may be simple units or their connection to the surface may be branched. Simple unbranched tubular glands exist in the walls of many of the hollow viscera, e.g. the small intestine and uterus, whereas some simple glands have expanded, flask-like ends (acini or alveoli). Such glands may consist entirely of secretory cells, or may have a blind-ending secretory portion that leads through a non-secretory duct to the surface, in which case the ducts may modify the secretions as they pass along them.

Glands with ducts may be branched (compound) and sometimes form elaborate ductal trees. Such glands generally have acinar or alveolar secretory lobules, as in the exocrine pancreas, but the secretory units may alternatively be tubular or mixed tubulo-acinar. More than one type of secretory cell may occur within a particular secretory unit, or individual units may be specialized to just one type of secretion (e.g. serous acini of salivary glands).

Exocrine glands are also classified by their secretory products. Secretory cells in mucus-secreting or mucous glands have frothy cytoplasm and basal, flattened nuclei. They stain deeply with metachromatic stains and periodic acid–Schiff (PAS) methods that detect carbohydrate residues. However, in general (i.e. non-specific) histological preparations, they are weakly stained because much of their content of water-rich mucin has been extracted by the processing procedures. Secretory cells in serous glands have centrally placed nuclei and eosinophilic secretory storage granules in their cytoplasm. They secrete mainly glycoproteins (including lysozyme) and digestive enzymes.

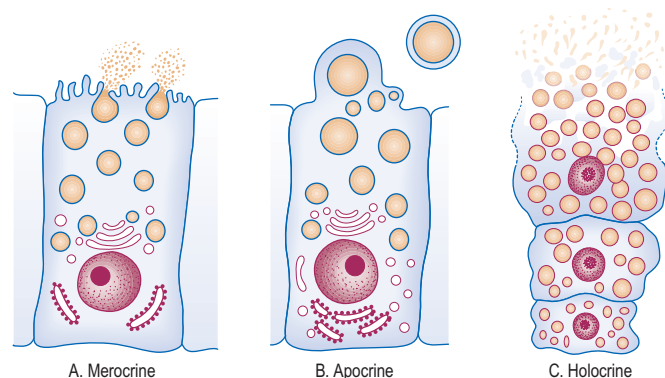
Some glands are almost entirely mucous (e.g. the sublingual salivary gland), whereas others are mainly serous (e.g. the parotid salivary gland). The submandibular gland is mixed, in that some lobules are predominantly mucous and others serous. Mucous acini may share a lumen with clusters of serous cells (seen in routine preparations as serous demilunes). Although this simple approach to classification is useful for general descriptive purposes, the diversity of molecules synthesized and secreted by glands is such that complex mixtures often exist within the same cell.

## ENDOCRINE GLANDS

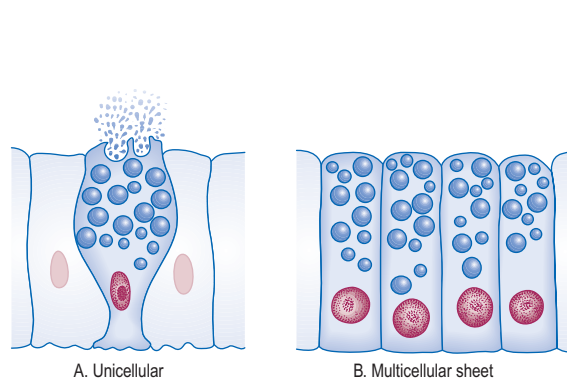
Endocrine glands secrete directly into connective tissue interstitial fluid and thence the circulation. Their cells are grouped around beds of capillaries or sinusoids, which typically are lined by fenestrated endothelia to allow the rapid passage of macromolecules through their walls. Endocrine cells may be arranged in clusters within vascular networks, in cords between parallel vascular channels or as hollow structures (follicles) surrounding their stored secretions. In addition to the cells of specialized ductless endocrine glands (e.g. pituitary, pineal, thyroid and parathyroid), hormone-producing cells also form components of other organ systems. These include: the cells of the pancreatic islets; thymic epithelial cells; renin-secreting cells of the kidney juxtaglomerular apparatus; erythropoietin-secreting cells of the kidney; circumventricular organs; interstitial testicular (Leydig) cells; interstitial follicular and luteal ovarian cells; and placental cells (in pregnancy). Some cardiac myocytes, particularly in the walls of the atria, also have endocrine functions. These cells are all described in detail within the appropriate regional sections.



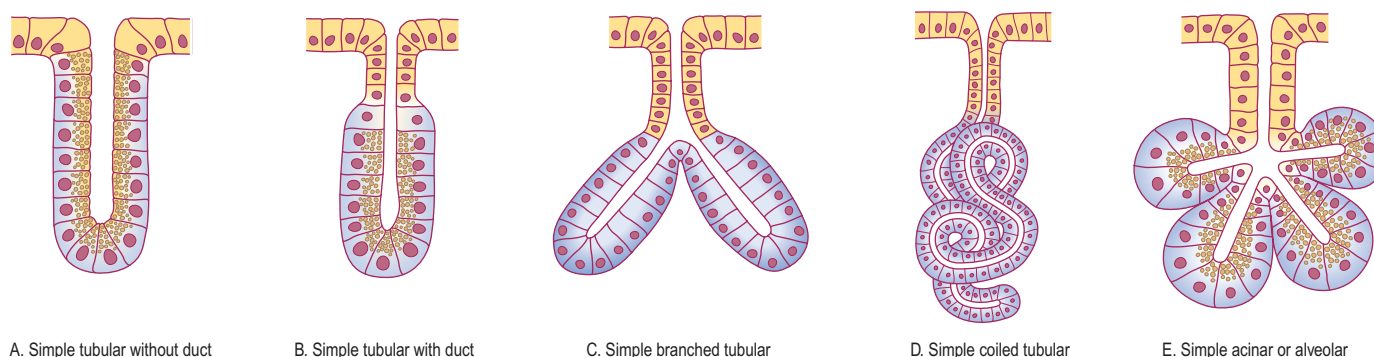
## Mechanisms of secretion



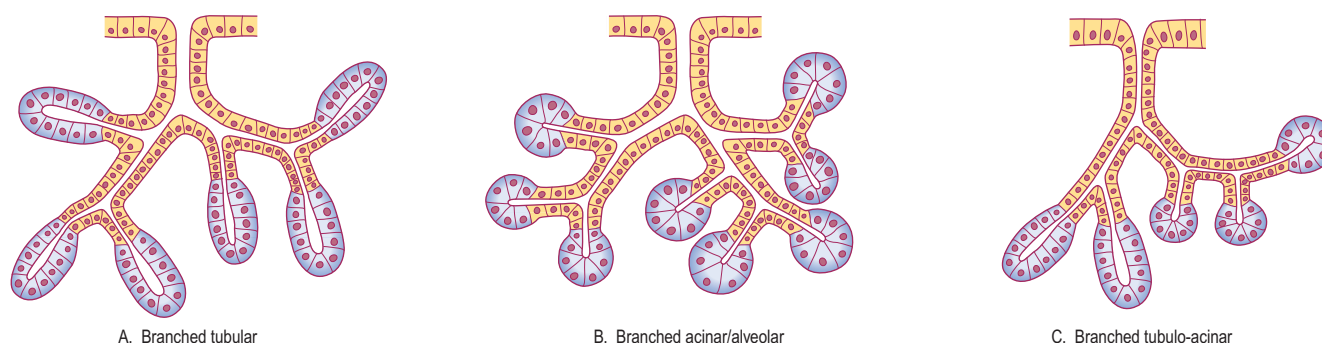
## Arrangement of cells



## Structural classification of glands – Simple glands with unbranched ducts



## Structural classification of glands – Ductal branching pattern of complex glands



**Fig. 2.6** Classification of the different types of epithelial gland.

Isolated endocrine cells also exist scattered amongst other tissues as part of the dispersed (diffuse) neuroendocrine system, e.g. throughout the alimentary and respiratory tracts. Neuroendocrine cells are generally situated within a mucosal epithelium and their bases often rest on the basal lamina (see below). In response to an external stimulus, they secrete their product basally into interstitial fluid. A typical neuroendocrine cell is shown in [Figure 2.7](#). The secretory granules vary in shape, size and ultrastructure according to cell type. Cells often take the name of the secretion they produce, e.g. gastrin-secreting G cells of the small intestine. Neuroendocrine cells share many of their secretory products with chemical mediators in the nervous system.

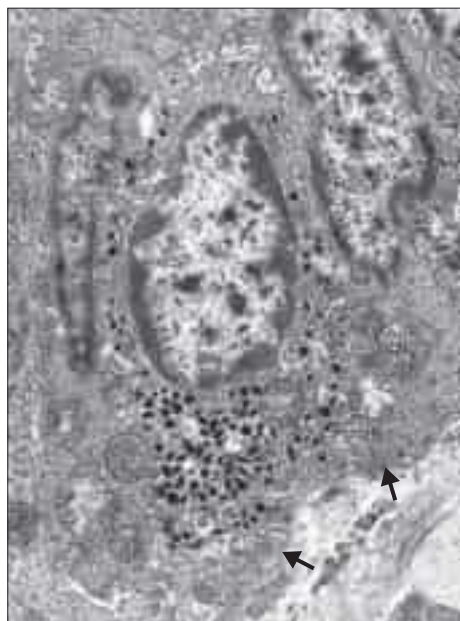
## CONTROL OF GLANDULAR SECRETION

The activities of cells in the various tissue and organ systems of the body are tightly regulated by the coordinated activity of the endocrine and autonomic nervous systems. Endocrine (and paracrine) signals reach target cells in interstitial fluid, often via blood plasma, and together with autonomic nervous signals they ensure that the body responds to normal physiological stimuli and adjusts to changes in the external environment. Hormone secretion is itself controlled in a number of ways, e.g. by neural control, regulatory feedback loops or according to various cyclical, rhythmical or pulsatile patterns of release. Endocrine

glands have a rich vascular supply and their blood flow is controlled by autonomic vasomotor nerves, which can thus modify glandular activity.

Glandular activity may also be controlled directly by autonomic secretomotor fibres, which may either form synapses on the bases of gland cells (e.g. in the suprarenal medulla) or release neuromediators in the vicinity of the glands and reach them by diffusion. Alternatively, the autonomic nervous system may act indirectly on gland cells, e.g. on neuroendocrine G cells via histamine, released neurogenically from another neuroendocrine cell in the gastric lining. Such paracrine activities of neuroendocrine cells are also important in the respiratory system.

Circulating hormones from the adenohypophysis stimulate synthesis and secretion by target cells in many endocrine glands. Such signals, mostly detected by receptors at the cell surface and mediated by second messenger systems, may increase the synthetic activity of gland cells, and may cause them to discharge their secretions by exocytosis. Secretions from certain exocrine glandular cells are expressed rapidly from those glands by the contraction of associated myoepithelial cells (see [Figs 2.3, 2.4](#)) that enclose the secretory units and smaller ducts. Myoepithelial cells may be under direct neural control, as in the salivary glands, or they may respond to circulating hormones, as in the mammary gland, where they respond to the concentration of circulating oxytocin.



**Fig. 2.7** An electron micrograph of a neuroendocrine cell between two absorptive cells in the colon (rat tissue). Dense neurosecretory granules are seen in the basal cytoplasm, apposed to the basal lamina (arrows). (Courtesy of Michael Crowder MD.)

## Feedback loops and endocrine axes

The pituitary gland, in particular the adenohypophysis, is often termed the master gland because of its central role in endocrine physiological processes. It provides the means by which the central nervous system regulates and integrates, by non-neural mechanisms, the widespread functions of the body, including the activities of other endocrine glands and, often indirectly, exocrine glands such as the breast. Regulatory hormones from the adenohypophysis stimulate synthesis and secretion in target cells of many endocrine glands; these glands therefore respond to, as well as generate, hormonal signals.

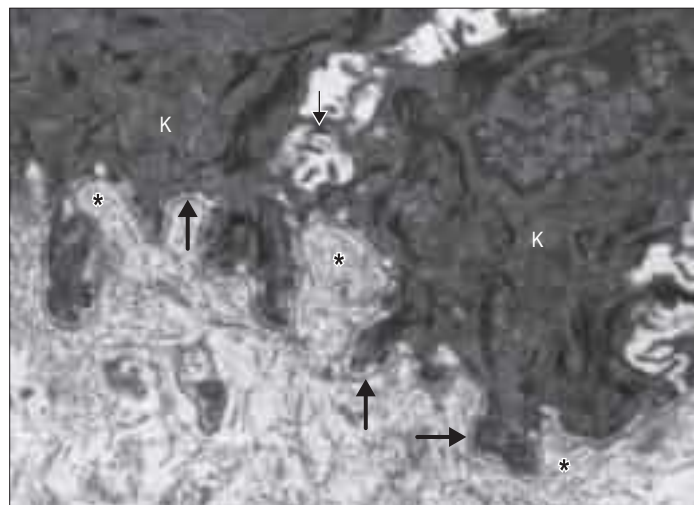
The hypothalamus and the adenohypophysis in the brain are central to most regulatory feedback loops within the endocrine system. Loops can be either positive or negative, e.g. the hypothalamus stimulates release of follicle stimulating hormone (FSH) by the adenohypophysis, which in turn stimulates ovarian follicular maturation and secretion of oestradiol, which acts on breast and endometrial target tissues. Oestradiol, in this case, also acts back on the adenohypophysis and hypothalamus to reinforce their function positively in a feedback loop. In contrast, hypothalamic and adenohypophysial stimulation of testicular production of testosterone, which acts on targets such as skeletal muscle, is negatively regulated in a feedback loop generated by circulating testosterone. Such negative feedback regulation is a widely utilized physiological mechanism.

## BASEMENT MEMBRANE AND BASAL LAMINA

There is a narrow layer of extracellular matrix, which stains strongly for carbohydrates, at the interface between connective and other tissues, e.g. between epithelia and their supporting connective tissues. In early histological texts this layer was termed the basement membrane. As almost all of its components are synthesized by the epithelium or other tissues, rather than the adjacent connective tissue, it will be discussed here.

Electron microscopy revealed that the basement membrane is composed of two distinct components. A thin, finely fibrillar layer, the basal lamina, is associated closely with the basal cell surface (Fig. 2.8). A variable reticular lamina of larger fibrils and glycosaminoglycans of the extracellular matrix underlies this layer and is continuous with the connective tissue proper, although it is much reduced or largely absent in some tissues, e.g. surrounding muscle fibres, Schwann cells and capillary endothelia. In other tissues, the basal lamina separates two layers of cells and there are no intervening typical connective tissue elements. This occurs in the thick basal lamina of the renal glomerular filter and the basal lamina of the thin portions of the lung interalveolar septa across which gases exchange between blood and air.

The basal lamina is usually about 80 nm thick, varying between 40 and 120 nm, and consists of a sheet-like fibrillar layer, the lamina densa (20–50 nm wide), separated from the plasma membrane of the cell it supports by a narrow electron-lucent zone, the lamina lucida. The lamina lucida is absent from tissues prepared by rapid freezing and so may be an artefact. In many tissues this zone is crossed by integral



**Fig. 2.8** The basal lamina as seen in an electron micrograph, underlying the basal epithelial layer of human skin (see Fig. 7.3). The finely fibrillar dense layer (long arrows) corresponds to the lamina densa, and fine collagen fibrils (\*) lie in the subjacent connective tissue. These contribute to the appearance of the basement membrane in light microscope preparations stained for carbohydrate-rich structures. The two cells seen in the upper field are basal keratinocytes (K), joined by desmosomes (short arrow), with dense keratin filaments in their cytoplasm. (Courtesy of J McMillan MD, St John's Institute of Dermatology, St Thomas' Hospital, London.)

plasma membrane proteins, e.g. keratinocyte hemidesmosomes are anchored into the lamina densa in the basal lamina of the epidermis. The basal lamina is a delicate felt-like network composed largely of two glycoprotein polymers, laminin and type IV collagen, which self-assemble into two-dimensional sheets interwoven with each other. Early embryonic basal lamina is formed only of the laminin polymer. Two other molecules cross-link and stabilize the network: entactin (nidogen) and perlecan (a large heparan sulphate proteoglycan).

Although all basal laminae have a similar form, their thickness and precise molecular composition vary between tissues and even within a tissue, e.g. between the crypts and villi of the small intestine. The isoforms of laminin and collagen type IV differ in various tissues; thus Schwann cells and muscle cells express laminin-2 (merosin) rather than the prototypical laminin-1. Laminin-5, although not itself a basal lamina component, is found in the hemidesmosomes of the basal epidermis and links the basal lamina with epidermal transmembrane proteins,  $\alpha_6\beta_4$  integrin and collagen type XVII (BPAG2, bullous pemphigoid antigen 2, one of the targets of the autoimmune blistering skin disease, bullous pemphigoid). The particular isoform of collagen type IV in the basal lamina of different tissues is reflected in tissue-specific disease patterns. Mutations in a collagen expressed by muscle and kidney glomeruli cause Alport's syndrome, a form of renal failure. Renal failure also occurs in Goodpasture's syndrome, in which renal basal lamina collagen is targeted by autoantibodies.

In Descemet's membrane in the cornea, collagen type VIII replaces collagen type IV in the much thickened (increasing with age, up to 10  $\mu\text{m}$ ,) endothelial basal lamina. The basal lamina of the neuromuscular junction contains agrin, a heparan sulphate proteoglycan, which plays a part in the clustering of muscle acetylcholine receptors in the plasma membrane at these junctions.

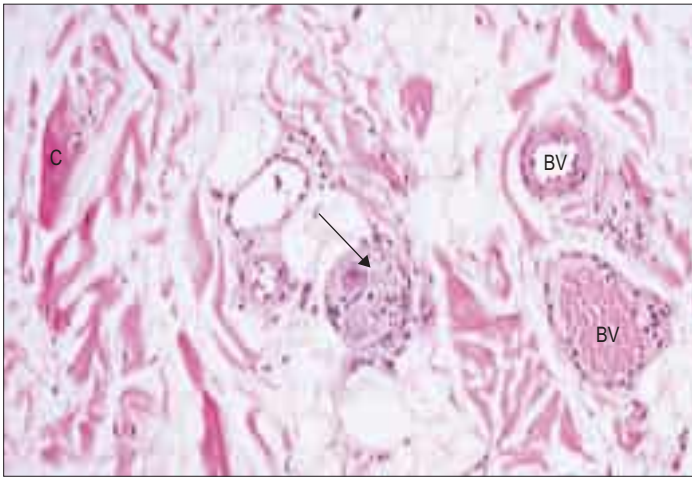
## RETICULAR LAMINA

The reticular lamina consists of a dense extracellular matrix that contains collagen. In skin, it contains fibrils of type VII collagen (anchoring fibrils), which bind the lamina densa to the adjacent connective tissue. The high concentration of proteoglycans in the reticular lamina is responsible for the positive reaction of the entire basement membrane to stains for carbohydrates, which is seen in sections prepared for light microscopy.

## FUNCTIONS OF BASAL LAMINA

Basal laminae perform a number of important roles (Iozzo 2005). They form selectively permeable barriers (anionic filters) between adjacent tissues, e.g. in the glomerular filter of the kidney; anchor epithelial and





**Fig. 2.9** General loose connective tissue (human), with bundles of collagen fibres (C) within an amorphous ground substance, penetrated by a neurovascular bundle of blood vessels (BV), lymphatics and nerves. A small autonomic ganglion is arrowed. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

connective tissues, and so stabilize and orientate the tissue layers; may exert instructive effects on adjacent tissues, and so determine their polarity, rate of cell division, cell survival, etc.; and regulate angiogenesis. In addition, they may act as pathways for the migration and pathfinding activities of growing cell processes, both in development and in tissue repair, e.g. in guiding the outgrowth of axons and the re-establishment of neuromuscular junctions during regeneration after injury in the peripheral nervous system. Changes in basal lamina thickness are often associated with pathological conditions, e.g. the thickening of the glomerular basal lamina in glomerulonephritis and diabetes.

## CONNECTIVE AND SUPPORTING TISSUES

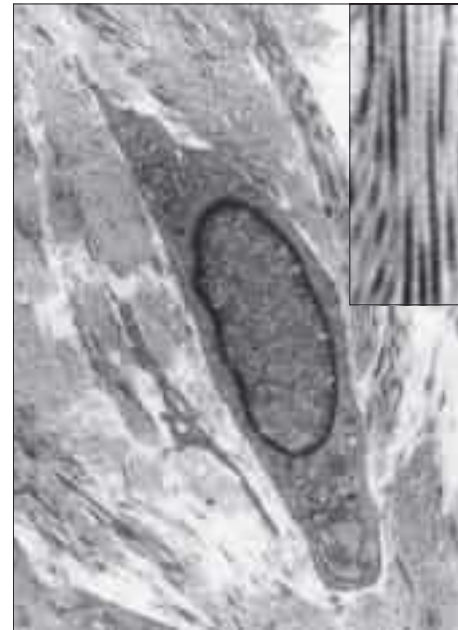
The connective tissues are defined as those composed predominantly of intercellular material, the extracellular matrix, which is secreted mainly by the connective tissue cells. The cells are therefore usually widely separated by their matrix, which is composed of fibrous proteins and a relatively amorphous ground substance (Fig. 2.9). Many of the special properties of connective tissues are determined by the composition of the matrix, and their classification is also largely based on its characteristics. In some types of connective tissue, the cellular component eventually dominates the tissue, even though the tissue originally has a high matrix:cell ratio, e.g. adipose tissue. Connective tissues are derived from embryonic mesenchyme or, in the head region, largely from neural crest.

Connective tissues have several essential roles in the body. These may be subdivided into structural roles, which largely reflect the special mechanical properties of the extracellular matrix components, and defensive roles, in which the cellular component has the dominant role. Connective tissues often also play important trophic and morphogenetic parts in organizing and influencing the growth and differentiation of surrounding tissues, e.g. in the development of glands from an epithelial surface.

Structural connective tissues are divided into ordinary (or general) types, which are widely distributed, and special skeletal types, i.e. cartilage and bone, which are described in Chapter 5. A third type, haemolymphoid tissues, consists of peripheral blood cells, lymphoid tissues and their precursors; these tissues are described in Chapter 4. They are often grouped with other types of connective tissue because of their similar mesenchymal origins and because the various defensive cells of the blood also form part of a typical connective tissue cell population. They reach connective tissues via the blood circulation and migrate into them through the endothelial walls of vessels.

## CELLS OF GENERAL CONNECTIVE TISSUES

Cells of general connective tissues can be separated into the resident cell population (fibroblasts, adipocytes, mesenchymal stem cells, etc.) and a population of migrant cells with various defensive functions (macrophages, lymphocytes, mast cells, neutrophils and eosinophils), which may change in number and moderate their activities according



**Fig. 2.10** An electron micrograph of a fibroblast in human connective tissue, surrounded by bundles of finely banded collagen fibrils (shown at high magnification in the insert), which they secrete. (Courtesy of Dr Bart Wagner, Histopathology Department, Sheffield Teaching Hospitals, UK.)

to demand. Embryologically, fibroblasts and adipocytes arise from mesenchymal stem cells, some of which may remain in the tissues to provide a source of replacement cells postnatally. As noted above, the cells of haemopoietic origin migrate into the tissue from bone marrow and lymphoid tissue.

## Resident cells

### Fibroblasts

Fibroblasts are usually the most numerous resident cells. They are flattened and irregular in outline, with extended processes, and in profile they appear fusiform or spindle-shaped (Fig. 2.10; see also Fig. 2.12). Fibroblasts synthesize most of the extracellular matrix of connective tissue (see Fig. 2.10); accordingly, they have all the features typical of cells active in the synthesis and secretion of proteins. Their nuclei are relatively large and euchromatic, and possess prominent nucleoli. In young, highly active cells, the cytoplasm is abundant and basophilic (reflecting the high concentration of rough endoplasmic reticulum), mitochondria are abundant and several sets of Golgi apparatus are present. In old and relatively inactive fibroblasts (often termed fibrocytes), the cytoplasmic volume is reduced, the endoplasmic reticulum is sparse and the nucleus is flattened and heterochromatic.

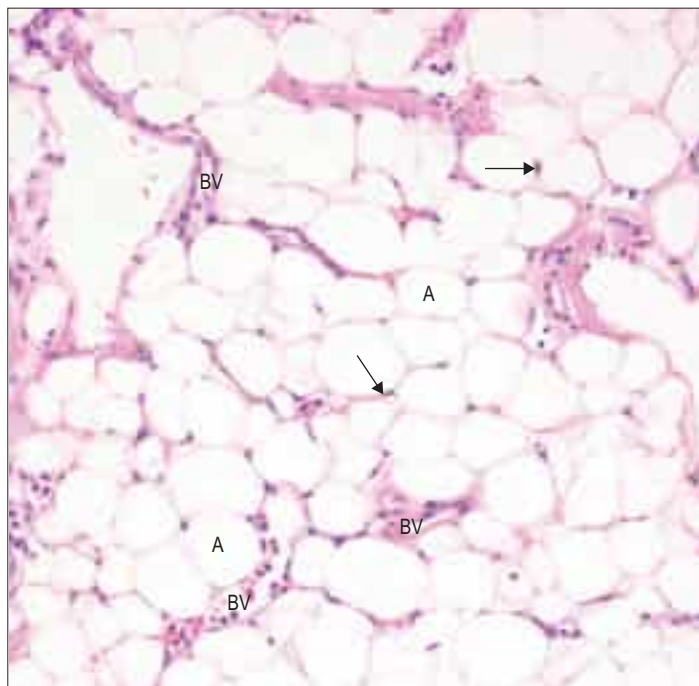
Fibroblasts are usually adherent to the fibres of the matrix (collagen and elastin), which they lay down. In some highly cellular structures, e.g. liver, kidney and spleen, and in most lymphoid tissue, fibroblasts and delicate collagenous fibres (type III collagen; reticular fibres) form fibrocellular networks, which are often called reticular tissue. The fibroblasts may then be termed reticular cells or reticulocytes.

Fibroblasts are particularly active during wound repair following traumatic injury or inflammation, when tissue mass is lost through cell death. They proliferate and lay down a fibrous matrix that becomes invaded by numerous blood vessels (granulation tissue). Contraction of wounds is, at least in part, caused by the shortening of myofibroblasts, specialized contractile fibroblast-like cells (Hinz et al 2012) with properties similar to smooth muscle cells. It was thought that myofibroblasts differentiated from fibroblasts (reviewed in McAnulty (2007)) or their progenitor mesenchymal stem cells (see below) in granulation tissue. However, recent evidence suggests that in wound healing and in many fibrotic disease processes, including hepatic cirrhosis, the myofibroblast precursor is the vascular pericyte or a closely related cell (reviewed in Duffield (2012)). In cases where the specialized cells of the damaged region cannot divide and regenerate functional tissue, e.g. cardiac muscle cells after infarction, connective tissue fibroblasts and their extracellular matrix fill the void to form a scar. An exception is the central nervous system, where glial scars are formed after injury. Fibroblast activity is influenced by various factors such as steroid hormone concentration, dietary content and prevalent mechanical stresses. Collagen formation is impaired in vitamin C deficiency.

### Adipocytes (lipocytes, fat cells)

Adipocytes occur singly or in groups in many, but not all, connective tissues. They are numerous in adipose tissue (Fig. 2.11). Individually,



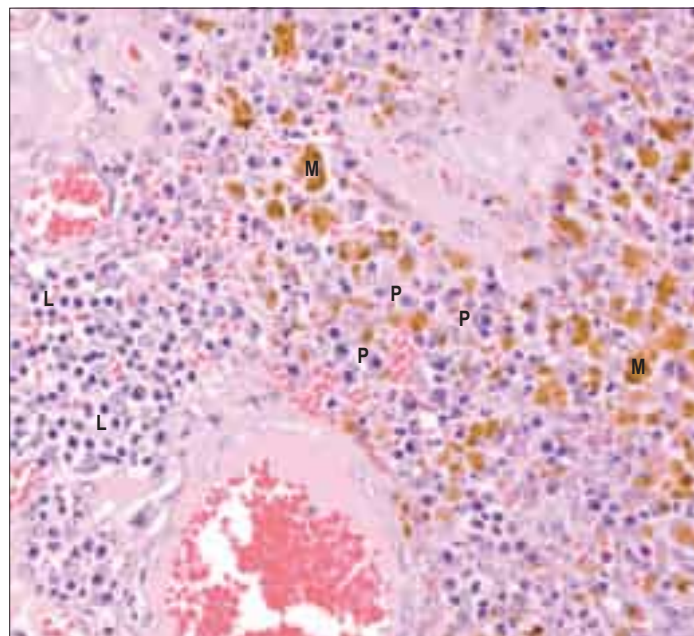


**Fig. 2.11** Adipose tissue (human, from a lymph node specimen). Adipocytes (A) are distended polygonal cells filled with lipid, which has been extracted by the tissue processing. This leaves only the plasma membranes with scant cytoplasm and nuclei (arrows), occasionally visible compressed against the cell periphery. Small blood vessels (BV) penetrate the adipose tissue; larger vessels are seen on the right. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

the cells are oval or spherical in shape, but when packed together they are polygonal. They vary in diameter, averaging 50  $\mu\text{m}$ . Each cell consists of a peripheral rim of cytoplasm, in which the nucleus is embedded, surrounding a single large central globule of fat, which consists of glycerol esters of oleic, palmitic and stearic acids. There is a small accumulation of cytoplasm around the oval nucleus, which is typically compressed against the cell membrane by the lipid droplet, together with the Golgi complex. Many cytoskeletal filaments, some endoplasmic reticulum and a few mitochondria lie around the lipid droplet, which is in direct contact with the surrounding cytoplasm and not enclosed within a membrane. In sections of tissue not specially treated to preserve lipids, the lipid droplet is usually dissolved out by the solvents used in routine preparations, so that only the nucleus and the peripheral rim of cytoplasm surrounding a central empty space remain.

Another form of adipose tissue, brown fat, occurs in the interscapular region of neonates, a location it shares with the classic brown fat of rodents. Brown fat is characterized by the presence of large cells, each of which contains several separate droplets of fat (multilocular adipose tissue) rather than a single globule (typical of unilocular adipose tissue; see above), and by mitochondria in which the cristae are unusually large and numerous. White fat cells are specialized to store chemical energy, whereas the physiological role of brown adipose tissue (BAT) cells is to metabolize fatty acids and generate heat; BAT cells uncouple cellular respiration via the mitochondrial uncoupling protein UCP1. It had been thought that brown fat disappears during postnatal growth, but significant deposits of UCP1-positive brown fat have been detected by positron emission tomography (PET) scanning methods in adults, mainly in the supraclavicular region, in the neck and along the spine. Recent evidence suggests that these human UCP1-positive cells may not be classic brown fat cells but a distinct type of thermogenic fat cell called a beige fat cell, thought to be derived from precursor cells in white fat (Wu et al 2012). Such cells may represent an evolutionarily conserved cellular mechanism to provide flexibility in adaptive thermogenesis.

It has long been recognized that adipose tissue is central to the control of energy balance and lipid homeostasis. There is a growing view that it may play a similarly important role as an endocrine organ, secreting a class of peptides called adipokines (Trayhurn and Wood 2004), which may enter the blood via capillaries or lymph. Different types of adipose tissue display functional and regional heterogeneity and differ in their involvement with disease processes (reviewed in Hassan et al (2012)). The mobilization of fat is under nervous or hormonal control; noradrenaline (norepinephrine) released at sympathetic



**Fig. 2.12** Macrophages (M) in chronically inflamed human connective tissue, showing prominent pigmented, haemosiderin-containing cytoplasmic granules derived from ingested erythrocytes. Many are multinucleate. Also seen are plasma cells (P), small lymphocytes (L) and other haemopoietic cells. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

nerve endings in adipose tissue is particularly important in this respect. No new adipose tissue is thought to form after the immediate postnatal period, and accumulation of body fat, as in obesity, is due to excessive accumulation of lipid in existing adipocytes, which become very large. Conversely, weight loss results from the mobilization and metabolism of lipid from adipocyte stores, with the consequent shrinkage of the cells.

### Mesenchymal stem cells

Mesenchymal stem cells are normally inconspicuous cells in connective tissues. They are derived from embryonic mesenchyme and are able to differentiate into the mature cells of connective tissue during normal growth and development, in the turnover of cells throughout life and, most conspicuously, in the repair of damaged tissues in wound healing. There is emerging evidence that, even in mature tissues, mesenchymal stem cells remain pluripotent and able to give rise to all the resident cells of connective tissues in response to local signals and cues. The potential therapeutic use of mesenchymal stem cell-based therapy for a wide range of autoimmune disorders and degenerative diseases is reflected in a burgeoning literature in the field of translational medicine. (See, for example, Ankrum and Karp (2010), Jackson et al (2012) and Ren et al (2012)).

### Migrant cells

#### Macrophages

Macrophages are numerous in connective tissues, where they are either attached to matrix fibres or are motile and migratory (Fig. 2.12). They are relatively large cells, 15–20  $\mu\text{m}$  in diameter, with indented and relatively heterochromatic nuclei and a prominent nucleolus. Their cytoplasm is slightly basophilic, contains many lysosomes and typically has a foamy appearance under the light microscope. Macrophages are important phagocytes and form part of the mononuclear phagocyte system. They can engulf and digest particulate organic materials, such as bacteria, and are able to clear dead or damaged cells from a tissue too. They are also the source of a number of secreted cytokines that have profound effects on many other cell types. Macrophages are able to proliferate in connective tissues to a limited extent, but are derived and replaced primarily from haemopoietic stem cells in the bone marrow, which circulate in the blood as monocytes before migrating through vessel walls into connective tissues, where they differentiate.

Many properties of macrophages in general connective tissue are similar to those of related cells in other sites. These include: circulating monocytes, from which they are derived; alveolar macrophages in the

lungs, which take up inhaled particles not cleared by the mucociliary rejection current; phagocytic cells in the lymph nodes, spleen and bone marrow; Kupffer cells of the liver sinusoids; and microglial cells of the central nervous system.

### Lymphocytes

Lymphocytes are normally present in small numbers; they are numerous in general connective tissue only in pathological states, when they migrate in from adjacent lymphoid tissue or from the circulation. The majority are small cells (6–8  $\mu\text{m}$ ) with highly heterochromatic nuclei but they enlarge when stimulated. Two major functional classes exist, termed B and T lymphocytes. B lymphocytes originate in the bone marrow, then migrate to various lymphoid tissues, where they proliferate. When antigenically stimulated, they undergo further mitotic divisions, then enlarge as they mature, commonly in general connective tissues, to form plasma cells that synthesize and secrete antibodies (immunoglobulins). Mature plasma cells are rounded or ovoid, up to 15  $\mu\text{m}$  across, and have an extensive rough endoplasmic reticulum. Their nuclei are spherical, often eccentrically situated, and have a characteristic 'clock-face' configuration of heterochromatin (see Fig. 4.12) that is regularly distributed in peripheral clumps. The prominent Golgi complex is visible with a light microscope as a pale region to one side of the nucleus and the remaining cytoplasm is deeply basophilic because of the abundant rough endoplasmic reticulum. Mature plasma cells do not divide.

T lymphocytes originate from precursors in bone marrow haemopoietic tissue but later migrate to the thymus, where they develop T-cell identity, before passing into the peripheral lymphoid system, where they continue to multiply. When antigenically stimulated, T cells enlarge and their cytoplasm becomes filled with free polysome clusters. The functions of T lymphocytes are numerous: different subsets recognize and destroy virus-infected cells, tissue and organ grafts, or interact with B lymphocytes and several other defensive cell types.

### Mast cells

Mast cells are important defensive cells. They occur particularly in loose connective tissues and in the fibrous capsules of certain organs such as the liver, and are numerous around blood vessels. Mast cells are round or oval, approximately 20  $\mu\text{m}$  in diameter, with many filopodia extending from the cell surface. The nucleus is centrally placed and relatively small. The cytoplasm contains large numbers of prominent vesicles and a well-developed Golgi apparatus, but scant endoplasmic reticulum. The vesicles have a high content of glycosaminoglycans and show a strongly positive reaction with the PAS stain for carbohydrates. They are membrane-bound, vary in size and shape (mean diameter 0.5  $\mu\text{m}$ ) and also have a rather heterogeneous content of dense, lipid-containing material, which may be finely granular, lamellar or in the form of membranous whorls.

The major granule components, many of them associated with inflammation (Frenzel and Hermine 2013), are the proteoglycan heparin, histamine, tryptase, superoxide dismutase, aryl sulphatase,  $\beta$ -hexosaminidase and various other enzymes, including chymase in connective tissue but not mucosal mast cells, together with chemotactic factors for neutrophil and eosinophil granulocytes. There are functional differences between mast cells found in different tissues.

Mast cells may be stimulated to release some or all of their contents, either by direct mechanical or chemical trauma, or after contact with particular antigens to which the body has previously been exposed. The consequences of granule release include alteration of capillary permeability, smooth muscle contraction, and activation and attraction to the locality of various other defensive cells. Responses to mast cell degranulation may be localized, e.g. urticaria, or there may occasionally be a generalized response to the release of large amounts of histamine into the circulation (anaphylactic shock). Mast cells closely resemble basophil granulocytes of the general circulation but are thought to develop as distinct descendants of an earlier myeloid lineage precursor. It is believed that they are generated in the bone marrow and circulate to the tissues as immature basophil-like cells, migrating through the capillary and venule walls to their final destination. For further reading, see Bischoff (2007) and Collington et al (2011).

### Granulocytes (polymorphonuclear leukocytes)

Neutrophil and eosinophil granulocytes are immigrant cells from the circulation. Relatively infrequent in normal connective tissues, their numbers may increase dramatically in infected tissues, where they are important components of cellular defence. Neutrophils are highly phagocytic, especially towards bacteria. The functions of eosinophils are less well understood. These cells are described further in Chapter 4.

## CELLS OF SPECIALIZED CONNECTIVE TISSUES

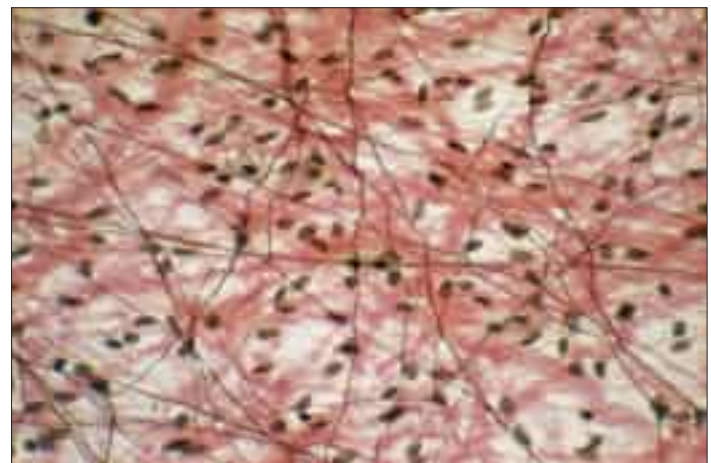
Skeletal tissues – namely, cartilage and bone – are generally classified with the connective tissues, but their structure and functions are highly specialized and they are described in Chapter 5. As with the general connective tissues, these specialized types are characterized by their extracellular matrix, which forms the major component of the tissues and is responsible for their properties. The resident cells are different from those in general connective tissues. Cartilage is populated by chondroblasts, which synthesize the matrix, and by mature chondrocytes. Bone matrix is elaborated by osteoblasts. Their mature progeny, osteocytes, are embedded within the matrix, which they help to mineralize, turn over and maintain. A third cell type, the osteoclast, has a different lineage origin and is derived from haemopoietic tissue; osteoclasts are responsible for bone degradation and remodelling in collaboration with osteoblasts.

## EXTRACELLULAR MATRIX

The term extracellular matrix is applied collectively to the extracellular components of connective and supporting tissues. Essentially, it consists of a system of insoluble protein fibres, adhesive glycoproteins and soluble complexes composed of carbohydrate polymers linked to protein molecules (proteoglycans and glycosaminoglycans), which bind water. The extracellular matrix distributes the mechanical stresses on tissues and also provides the structural environment of the cells embedded in it, forming a framework to which they adhere and on which they can move (reviewed in Even-Ram and Yamada (2005) and Wolf and Friedl (2011)). With the exception of bone matrix, it provides a highly hydrated medium, through which metabolites, gases and nutrients can diffuse freely between cells and the blood vessels traversing it or, in the case of cartilage, passing nearby.

There are many complex interactions between connective tissue cells and the extracellular matrix. The cells continually synthesize, secrete, modify and degrade extracellular matrix components, and respond to contact with the matrix in the regulation of cell metabolism, proliferation and motility. Degradation of the matrix is an important feature of embryonic development, morphogenesis, angiogenesis, tissue repair and remodelling (Mott and Werb 2004). Various types of proteinase are involved, principally metalloproteinases such as matrix metalloproteinases (MMPs), and those with a disintegrin and metalloproteinase domain (ADAMs) that include ADAMs with a thrombospondin domain (ADAMTS). Tissue remodelling depends on the controlled degradation of the extracellular matrix by secreted MMPs, regulated by their specific inhibitors, as occurs, for instance, during involution of the postpartum uterus or during menstrual lysis and shedding of the endometrium (Gaide Chevrionnay et al 2012). In the process of matrix degradation, bioactive peptides are liberated that act as growth factors, cytokines and other signalling molecules to change the behaviour of cells in the vicinity. While precisely regulated under physiological conditions, pathologically dysregulated extracellular matrix degradation is a cause of many diseases, such as atherosclerosis, emphysema, osteoarthritis and diabetic vascular complications.

The insoluble fibres are mainly of two types of structural protein: members of the collagen family, and elastin (Fig. 2.13). The



**Fig. 2.13** Elastic fibres, seen as fine, dark, relatively straight fibres in a whole-mount preparation of mesentery, stained for elastin. The wavy pink bands are collagen bundles and oval grey nuclei are mainly of fibroblasts.



## INTEGRATING CELLS INTO TISSUES

interfibrillar matrix (ground substance) includes a number of adhesive glycoproteins that perform a variety of functions in connective tissues, including cell–matrix adhesion and matrix–cell signalling. These glycoproteins include fibronectin, laminin, tenascin and vitronectin, in addition to a number of other less well characterized proteins. The glycosaminoglycans of the interfibrillar matrix are, with one notable exception, post-translationally modified proteoglycan molecules in which long polysaccharide side chains are added to short core proteins during transit through the secretory pathway between the rough endoplasmic reticulum and the *trans*-Golgi network. The exception, the polymeric disaccharide, hyaluronan, has no protein core and is synthesized entirely by cell surface enzymes. For further reading on extracellular matrix molecules, see Pollard et al (2008). Functional attributes of connective tissues vary and depend on the abundance of its different components. Collagen fibres resist tension, whereas elastin provides a measure of resilience to deformation by stretching. The highly hydrated, soluble polymers of the interfibrillar material (proteoglycans and glycosaminoglycans, mainly hyaluronan) generally form a stiff gel resisting compressive forces. Thus tissues that are specialized to resist tensile forces (e.g. tendons) are rich in collagen fibrils; tissues that accommodate changes in shape and volume (e.g. mesenteries) are rich in elastic fibres; and those that absorb compressive forces (e.g. cartilages) are rich in glycosaminoglycans and proteoglycans. In bone, mineral crystals take the place of most of the soluble polymers and endow the tissue with incompressible rigidity.

### Fibrillar matrix

#### Collagens

Collagens make up a very large proportion (approximately 30%) of all the proteins of the body. They consist of a wide range of related molecules that have various roles in the organization and properties of connective (and some other) tissues. The first collagen to be characterized was type I, the most abundant of all the collagens and a constituent of the dermis, fascia, bone, tendon, ligaments, blood vessels and the sclera of the eyeball. The characteristic collagen of cartilage and the vitreous body of the eye, with a slightly different chemical composition, is type II, whereas type III is present in several tissues, including the dermis and blood vessels, and type IV is in basal lamina. The other types are widely distributed in various tissues. Five of the collagens, types I, II, III, V and XI, form fibrils; types IV, VIII and X form sheets or meshworks; types VI, VII, IX, XII, XIV and XVIII have an anchoring or linking role; and types XIII and XVII are transmembrane proteins.

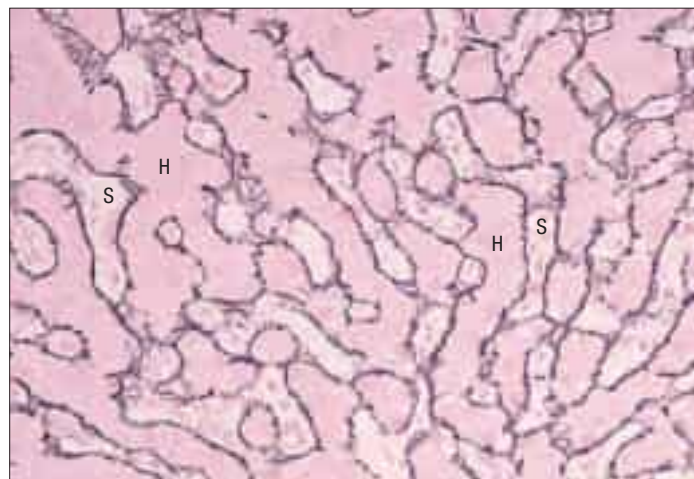
Biochemically, all collagens have a number of features in common. Unlike most other proteins, they contain high levels of hydroxyproline and all are composed of three polypeptides that form triple helices and are substantially modified post-translationally. After secretion, individual molecules are further cross-linked to form stable polymers. Functionally, collagens are structural proteins with considerable mechanical strength. Just a few of their distinguishing structural features are described below. For further reading on the molecular structure and functions of the collagens, see Pollard et al (2008).

#### Type I collagen

Type I collagen is very widely distributed. It forms inextensible fibrils in which collagen molecules (triple helices) are aligned side by side in a staggered fashion, with three-quarters of the length of each molecule in contact with neighbouring molecules. The fibril has well-marked bands of charged and uncharged amino acids arranged across it; these stain with heavy metals in a banding pattern that repeats every 65 nm in longitudinal sections viewed in the electron microscope (see Fig. 2.10 insert).

Fibril diameters vary between tissues and with age. Developing tissues often have thinner fibrils than mature tissues. Corneal stroma fibrils are of uniform and thin diameter, whereas tendon fibrils may be up to 20 times thicker and quite variable. Tissues in which the fibrils are subject to high tensile loading tend to have thicker fibrils. Thick fibrils are composites of uniform thin fibrils with a diameter of 8–12 nm. The fibrils themselves are relatively flexible, but when mineralized (as in bone) or surrounded by high concentrations of proteoglycan (as in cartilage), the resulting fibre-reinforced composite materials are rigid.

Fresh type I collagen fibres are tightly packed assemblies of parallel fibrils and are white and glistening. They form variably wavy (crimped) bundles of various sizes that are generally visible at the light microscope level. The component fibres may leave one bundle and interweave with others. In some situations, collagen fibrils are laid down in precise geometrical patterns, in which successive layers alternate in direction, e.g. corneal stroma, where the high degree of order is essential for transpar-



**Fig. 2.14** Reticular fibres (type III collagen; reticulin demonstrated by silver-staining) in human liver, forming a delicate meshwork within the space of Disse between hepatocytes (H), plasma membranes and the sinusoidal endothelia (S). (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

ency. Tendons, aponeuroses and ligaments are also highly ordered tissues (Ch. 5).

#### Types II, III, V and XI collagens

Types II, III, V and XI collagens can also aggregate to form linear fibrils. Type II collagen occurs in extremely thin (10 nm), short fibrils in the vitreous humour and in very thick fibrils in ageing human cartilage. The amino-acid sequence and banding pattern are very similar to those of type I collagen, as are the post-translational modifications of the triple helical protein molecule. The fine fibrils in the vitreous may fuse into thicker aggregates in older tissue.

Type III collagen is very widely distributed, particularly in young and repairing tissues. It usually co-localizes with type I collagen, and covalent links between type I and type III collagen have been demonstrated. In skin, many fibrils are probably composites of type I and type III collagens.

#### Reticular fibres

Fine branching and anastomosing reticular fibres form the supporting mesh framework of many glands, including the liver (Fig. 2.14), the kidney and lymphoreticular tissue (lymph nodes, spleen, etc.). Classically, these fibres stained intensely with silver salts, although they are poorly stained using conventional histological techniques. They associate with basal laminae and are often found in the neighbourhood of collagen fibre bundles. Reticular fibres are formed principally of type III collagen.

#### Elastin

Elastin is a 70 kDa protein, rich in the hydrophobic amino acids valine and alanine. Elastic fibrils, which also contain fibrillin, are highly cross-linked via two elastin-specific amino acids, desmosine and isodesmosine, which are formed extracellularly from lysine residues. They are less widely distributed than collagen, yellowish in colour, typically cross-linked and usually thinner (10–20 nm) than collagen fibrils. They can be thick, e.g. in the ligamenta flava and ligamentum nuchae. Unlike collagen type I, they show no banding pattern in the electron microscope. They stain poorly with routine histological stains but are stained with orcein-containing preparations (see Fig. 2.13). They sometimes appear as sheets, as in the fenestrated elastic lamellae of the aortic wall. Elastin-rich structures stretch easily with almost perfect recoil, although they tend to calcify with age and lose elasticity. Elastin is highly resistant to attack by acid and alkali, even at high temperatures.

### Interfibrillar matrix

#### Glycosaminoglycans

The structural soluble polymers characteristic of the extracellular matrix are the acidic glycosaminoglycans, which are unbranched chains of repeating disaccharide units, each unit carrying one or more negatively charged groups (carboxylate or sulphate esters, or both). The anionic charge is balanced by cations ( $\text{Na}^+$ ,  $\text{K}^+$ , etc.) in the interstitial fluid. Their polyanionic character endows the glycosaminoglycans with high osmotic activity, which helps to keep the fibrils apart, confers stiffness

on the porous gel that they collectively create, and gives the tissue a varying degree of basophilia. Glycosaminoglycans are named according to the tissues in which they were first found, e.g. hyaluronan (vitreal body), chondroitins (cartilage), dermatan (skin), keratan (cornea), heparan (liver). This terminology is no longer relevant, as most glycosaminoglycans are very widely distributed, whereas, conversely, some corneas contain little or no keratan sulphate. Of the glycosaminoglycans, all except hyaluronan have short protein cores and are highly variable in their carbohydrate side-chain structure.

### Hyaluronan

Hyaluronan was formerly called hyaluronic acid (or hyaluronate, as only the salt exists at physiological pH). It is a very large, highly hydrated molecule (25,000 kDa). Hyaluronan is found in all extracellular matrices and in most tissues, and is a prominent component of embryonic and developing tissues.

Hyaluronan is important in the aggregation of proteoglycans and link proteins that possess specific hyaluronan binding sites (e.g. laminin). Indeed, the very large aggregates that are formed may be the essential compression-resisting units in cartilage. Hyaluronan also forms very viscous solutions, which are probably the major lubricants in synovial joints. Because of its ability to bind water, it is often present in semi-rigid structures (e.g. vitreal humour in the eye), where it cooperates with sparse but regular meshworks of thin collagen fibrils.

### Proteoglycans

Proteoglycans have been classified according to the size of their protein core; their nomenclature is under review. The same core protein can bear different glycosaminoglycan side chains in different tissues. The functions of many proteoglycans are poorly understood. Some of the better-known proteoglycans are: aggrecan in cartilage, perlecan in basal laminae, decorin associated with fibroblasts in collagen fibril assembly, and syndecan in embryonic tissues.

### Adhesive glycoproteins

These proteins include molecules that mediate adhesion between cells and the extracellular matrix, often in association with collagens, proteoglycans or other matrix components. All of them are glycosylated and they are, therefore, glycoproteins. General connective tissue contains the well-known families of fibronectins (and osteonectin in bone), laminins and tenascins; there is a rapidly growing list of other glycoproteins associated with extracellular adhesion (Pollard et al 2008). They possess binding sites for other extracellular matrix molecules and for cell adhesion molecules, especially the integrins; in this way they enable cells selectively to adhere to and migrate through, appropriate matrix structures (reviewed in Jacquemet et al (2013)). They also function as signalling molecules, which are detected by cell surface receptors and initiate changes within the cytoplasm (e.g. to promote the formation of hemidesmosomes or other areas of strong adhesion; reorganize the cytoskeleton; and promote or inhibit locomotion and cell division).

### Fibronectin

Fibronectin is a large glycoprotein consisting of a dimer joined by disulphide links. Each subunit is composed of a string of large repetitive domains linked by flexible regions. Fibronectin subunits have binding sites for collagen, heparin and cell surface receptors, especially integrins, and so can promote adhesion between all these elements. In connective tissues, the molecules are able to bind to cell surfaces in an orderly fashion, to form short fibronectin filaments. The liver secretes a related protein, plasma fibronectin, into the circulation. The selective adhesion of different cell types to the matrix during development and in postnatal life is mediated by numerous isoforms of fibronectin generated by alternative splicing. Isoforms found in embryonic tissues are also expressed during wound repair, when they facilitate tissue proliferation and cell movements; the adult form is re-expressed once repair is complete.

### Laminin

Laminin is a large (850 kDa) flexible molecule composed of three polypeptide chains (designated  $\alpha$ ,  $\beta$  and  $\gamma$ ). There are many isoforms of the different chains, and at least 18 types of laminin. The prototypical molecule has a cruciform shape, in which the terminal two-thirds are wound round each other to form the stem of a cross, and the shorter free ends form the upright and transverse members. Laminin bears binding sites for other extracellular matrix molecules such as heparan sulphate, type IV collagen and entactin, and also for laminin receptor molecules (integrins) situated in cell plasma membranes. Laminin molecules can assemble themselves into flat regular meshworks, e.g. in the basal lamina.

### Tenascin

Tenascin is large glycoprotein composed of six subunits that are joined at one end to form a structure that resembles the spokes of a wheel. There is a family of tenascin molecules, generated by alternative splicing of the tenascin gene transcript. Tenascin is abundant in embryonic tissues but its distribution is restricted in the adult. It appears to be important in guiding cell migration and axonal growth in early development: it may either promote or inhibit these activities, depending on the cell type and tenascin isoform.

## CLASSIFICATION OF CONNECTIVE TISSUES

Connective and supporting tissues differ considerably in appearance, consistency and composition in different regions. These differences reflect local functional requirements and are related to the predominance of the cell types; the concentration, arrangement and types of fibre; and the characteristics of the interfibrillar matrix. On these bases, general connective tissues can be classified into irregular and regular types, according to the degree of orientation of their fibrous components.

### Irregular connective tissues

Irregular connective tissues can be further subdivided into loose, dense and adipose connective tissue.

#### Loose (areolar) connective tissue

Loose connective tissue is the most generalized form and is extensively distributed. Its chief function is to bind structures together, while still allowing a considerable amount of movement to take place. It constitutes the submucosa in the digestive tract and other viscera lined by mucosae, and the subcutaneous tissue in regions where this is devoid of fat (e.g. eyelids, penis, scrotum and labia), and it connects muscles, vessels and nerves with surrounding structures. It is present in the interior of organs, where it binds together the lobes and lobules of glands, forms the supporting layer (lamina propria) of mucosal epithelia and vascular endothelia, and lies within and between fascicles of muscle and nerve fibres.

Loose connective tissue consists of a meshwork of thin collagen and elastin fibres interlacing in all directions (see Fig. 2.13) to give a measure of both elasticity and tensile strength. The large meshes contain the soft, semi-fluid interfibrillar matrix or ground substance, and different connective tissue cells, which are scattered along the fibres or in the meshes. It also contains adipocytes, usually in small groups, and particularly around blood vessels.

A variant of loose connective tissue occurs in the choroid and the sclera of the eye, where large numbers of pigment cells (melanocytes) are also present.

#### Dense irregular connective tissue

Dense irregular connective tissue is found in regions that are under considerable mechanical stress and where protection is given to ensheathed organs. The matrix is relatively acellular and contains a high proportion of collagen fibres organized into thick bundles interweaving in three dimensions and imparting considerable strength. There are few active fibroblasts, which are usually flattened with heterochromatic nuclei. Dense irregular connective tissue occurs in: the reticular layer of the dermis; the superficial connective tissue sheaths of muscle and nerves, and the adventitia of large blood vessels; and the capsules of various glands and organs (e.g. testis, sclera of the eye, periosteum and perichondria).

#### Adipose tissue

A few adipocytes occur in loose connective tissue in most parts of the body. However, they constitute the principal component of adipose tissue (see Fig. 2.11), where they are embedded in a vascular loose connective tissue, usually divided into lobules by stronger fibrous septa carrying the larger blood vessels. Adipose tissue only occurs in certain regions. In particular it is found: in subcutaneous tissue; in the mesenteries and omenta; in the female breast; in bone marrow; as retro-orbital fat behind the eyeball; around the kidneys; deep to the plantar skin of the foot; and as localized pads in the synovial membrane of many joints. Its distribution in subcutaneous tissue shows characteristic age and sex differences. Fat deposits serve as energy stores, sources of metabolic lipids, thermal insulation (subcutaneous fat) and mechanical shock-absorbers (e.g. soles of the feet, palms of the hands, gluteal region and synovial membranes).





**Fig. 2.15** Dense regular connective tissue in a tendon. Thick parallel bundles of type I collagen (here stained pink) give tendon its white colour in life. The elongated nuclei of inactive fibroblasts (tendon cells) are visible between collagen bundles.

## Regular connective tissues

Regular connective tissues include highly fibrous tissues in which fibres are regularly orientated, either to form sheets such as fasciae and aponeuroses, or as thicker bundles such as ligaments or tendons (Fig. 2.15). The direction of the fibres within these structures is related to the stresses that they undergo: fibrous bundles display considerable interweaving, even within tendons, which increases their structural stability and resilience.

The fibroblasts that secrete the fibres may eventually become trapped within the fibrous structure, where they become compressed, relatively inactive cells with stellate profiles and small heterochromatic nuclei; these cells are called tendon cells. Fibroblasts on the external surface may be active in continued fibre formation and they constitute a pool of cells available for repair of injured tissue.

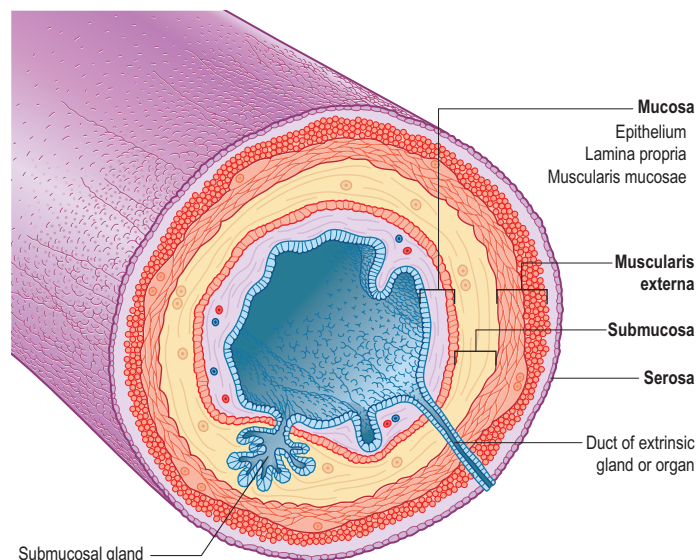
Although regular connective tissue is predominantly collagenous, some ligaments contain significant amounts of elastin, e.g. the ligamenta flava of the vertebral laminae and the vocal folds. The collagen fibres may form precise geometrical patterns, as in the cornea.

## Mucoid tissue

Mucoid tissue is found chiefly as a stage in the development of connective tissue from mesenchyme. It exists in Wharton's jelly, which forms the bulk of the umbilical cord, and consists substantially of extracellular matrix, largely made up of hydrated mucoid material and a fine meshwork of collagen fibres, in which nucleated, fibroblast-like cells with branching processes are found. Fibres are usually rare in typical mucoid tissue, although the full-term umbilical cord contains perivascular collagen fibres. Postnatally, mucoid tissue is seen in the pulp of a developing tooth, the vitreous body of the eye (a persistent form of mucoid tissue that contains few fibres or cells) and the nucleus pulposus of the intervertebral disc.

## TRANSDIFFERENTIATION AND METAPLASIA

Transitions occur between populations of cells forming an epithelium (sheets of polarized cells) and mesenchymal types (where the cells lack polarity) during normal development (see Thiery et al (2009)). In post-natal life, most well-described transitions between morphologically different cell types do not cross an epithelial–mesenchymal boundary but are transitions between types of epithelial cell or, less frequently, between mesenchymal (connective tissue) cell types. Most instances of such transdifferentiation (metaplasia, see Commentary 1.4) are adaptive, to changing environmental conditions or trauma, and almost all are pathological; the altered cells are termed metaplastic. A very common and physiologically normal example is the squamous metaplasia of columnar secretory epithelium of the distal endocervical canal, when exposed to the hormonally stimulated vaginal environment. Gastric metaplasia of the lower oesophagus may occur when chronic reflux of gastric juices exposes its stratified squamous epithelial lining to acid, and the original epithelium is replaced by a mucus-secreting columnar epithelium typical of the stomach (Barrett's oesophagus); this is pathological and susceptible to malignant change. Similarly, the respiratory epithelium (see Fig. 2.2D) of the upper airway often develops



**Fig. 2.16** A generalized mucosa and supporting tissues. For details and variations, see text.

foci of stratified squamous metaplasia in response to irritants in cigarette smoke. Mesenchymal (osseous) metaplasia can occur, for example, in the fibrous connective tissue of muscles subjected to repeated damage, where trabeculi of bone develop. It is thought that stem cells (rather than the differentiated cells) in the affected tissue respond to changes in their environment by altering their differentiation pathway, a process that may be reversible if the stimulus is removed.

## MUCOSA (MUCOUS MEMBRANE)

A mucosa or mucous membrane (Fig 2.16) lines many internal hollow organs in which the inner surfaces are moistened by mucus, such as the intestines, conducting portions of the airway, and the genital and urinary tracts. A mucosa proper consists of an epithelial lining, which may have the ducts of mucosal, submucosal or extrinsic glands opening on to its surface, an underlying loose connective tissue, the lamina propria, and a thin layer of smooth muscle, the muscularis mucosae. This last layer either may be absent from some mucosae, or may be replaced by a layer of elastic fibres. The term mucous membrane reflects the fact that these tissues can all be peeled away as a sheet or membrane from underlying structures; the plane of separation occurs along the muscularis mucosae.

Submucosa is a layer of supporting connective tissue that usually lies below the muscularis mucosae. It may contain mucous or seromucous submucosal glands. Inflammation of the viscera involves, primarily, the connective tissues of the submucosa and lamina propria, and is characterised by dilated vessels, oedema, and accumulations of extravasated immune defence cells. Most mucosae are also supported by one or more layers of smooth muscle, the muscularis externa. Contraction of this muscle may constrict the mucosal lumen (e.g. in the airway) or, where there are two or more muscle layers orientated in opposing directions (e.g. in the intestines), cause peristaltic movement of the viscus and the contents of its lumen. The outer surface of the muscle may be covered by a serosa or, where the structure is retroperitoneal or passes through the pelvic floor, by a connective tissue adventitia.

## MUCUS

Mucus is a viscous suspension of complex glycoproteins (mucins) of various kinds, and is secreted by scattered individual epithelial (goblet) cells, a secretory surface epithelium (e.g. the stomach lining) or mucous and seromucous glands. The precise composition of the mucus varies with the tissue and secretory cells that produce it. All mucins consist of filamentous core proteins to which are attached carbohydrate chains, usually branched; salivary mucus contains nearly 600 chains. Carbohydrate residues include glucose, fucose, galactose and *N*-acetylglucosamine (sialic acid). The terminals of some carbohydrate chains are identical to the blood group antigens of the ABO group in the majority of the population (secretors, bearing the secretor gene *S*<sup>c</sup>), and can be detected in salivary mucus by means of appropriate clinical tests. The long polymeric carbohydrate chains bind water and protect surfaces against

drying; they also provide good lubricating properties. In concentrated form, mucins form viscous layers that protect the underlying tissues against damage.

Synthesis of mucus starts in the rough endoplasmic reticulum. It is then passed to the Golgi complex, where it is conjugated with sulphated carbohydrates to form the glycoprotein, mucinogen, and this is exported in small, dense, membrane-bound vesicles that swell as they approach the cell surface, with which they fuse before releasing their contents.

## SEROSA (SEROUS MEMBRANE)

Serosa consists of a single layer of squamous mesothelial cells, expressing keratin intermediate filaments, supported by an underlying layer of loose connective tissue that contains numerous blood and lymphatic vessels. Serosa lines the pleural, pericardial and peritoneal cavities, and covers the external surfaces of organs lying within those cavities and, in the abdomen, the mesenteries that envelop them. A potential space, filled with a small amount of protein-containing serous fluid – largely an exudate of interstitial fluid – exists between the outer parietal and the inner visceral layers of the serosa.

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## FASCIA

Fascia is a generic term applied to sheaths, sheets or other dissectible masses of connective tissue that are large enough to be visible to the unaided eye. The terms superficial fascia and deep fascia, widely used to describe the connective tissue between the skin and underlying muscle, and the connective tissue surrounding muscles, viscera and related structures, respectively, are no longer included in the *Terminologia Anatomica*, although they remain in common usage in the English language. Tela subcutanea, hypodermis and subcutaneous tissue are the recommended synonymous terms that replace superficial fascia. Deeper-lying condensations of connective tissue have been defined according to their location, e.g. investing muscles (fascia musculorum) or viscera (fascia visceralis). Loosely packed connective tissue surrounds peripheral nerves, blood and lymph vessels as they pass between other structures, often linking them together as neurovascular bundles. Some large vessels, e.g. the common carotid and femoral arteries, are invested by a dense connective tissue sheath that may be functionally significant, aiding venous return by approximating large veins to pulsating arteries.



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